

Observer-Aware Topological Memory with Conserved Failed Reasoning: A Corrective Experimental Test of an SToE-Derived AI Architecture

Mykola Voronin

Faculty of Social Sciences and Social Technologies

Department of Psychology and Pedagogics

National University of “Kyiv-Mohyla Academy”, Kyiv, 04070, Ukraine

voronin.mykola@ukma.edu.ua

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Abstract

The Special Theory of Everything (SToE) has previously been used to derive a specific architectural prediction for artificial intelligence: reasoning systems should conserve failed hypotheses and their relational history in a persistent field, and should navigate that field according to structural connection rather than semantic similarity alone. Earlier SToE experiments using an ontology-priming bridge on generic logic puzzles produced null structural effects, motivating a change of experimental instrument rather than a stronger claim. This study tests a narrower mechanism: whether observer/current-state-aware topological navigation can recover previously rejected hypotheses when a later state change makes them relevant again. The v9.1 system stores canonical SToE seed information and runtime reasoning artifacts in one directed field, represents current observer state and state changes as information points, and ranks structurally reachable candidates using current-goal and provenance information. Eighteen tasks from nine families were evaluated under 14 frozen conditions with a local Gemma4:26b model, yielding 252 primary cells. On nine topology-targeted tasks, observer-aware topology solved 9/9, compared with 6/9 for dense semantic retrieval and 5/9 for BM25. The observer-minus-dense difference was +33.3 percentage points (family-cluster bootstrap 95% CI [0.0, 66.7], exact paired $p=.25$). Observer-aware topology also solved 9/9 targeted tasks versus 0/9 for query-blind topology and 0/9 when failed artifacts were removed; randomized edges solved 3/9. In all three observer-only wins over dense retrieval, removing the focal failed information point or the rejected-by relation class changed success to failure. V9.1 is a corrective replication: a prior v9 run was invalidated after discovering a 5,000-character serializer defect that produced unequal model-visible memory. The corrected run verified four visible artifacts in all 234 memory-bearing cells. The results provide preliminary support for a narrow SToE-derived architectural mechanism, not for the full ontology, and require independent replication on a new blind holdout benchmark.

Keywords: Special Theory of Everything; persistent reasoning graph; topology-aware memory; failed reasoning; observer-aware navigation; retrieval; corrective replication; AI architecture

1. Introduction

Contemporary language-model agents increasingly rely on external memory, retrieval, and self-evaluation mechanisms, but most retrieval pipelines still prioritize content similarity. That strategy is effective when the information needed now resembles the current query. It is less obviously suited to a different class of problem: a hypothesis was considered earlier, rejected for a specific reason, and later becomes useful because the state of the problem has changed. In such cases, the relevant old information may remain semantically distant from the current query even though it is structurally adjacent through the history of why it was rejected.

This problem is central to the AI implications derived from the Special Theory of Everything (SToE). The earliest SToE work treated information points and their connections as a general substrate for idea generation and artificial intelligence (Voronin, 2021). Later work formalized autoinjection as the framework’s self-containing mapping primitive (Voronin, 2025a), developed SToE as a cognitive architecture (Voronin, 2025b), stated the three foundational laws of information conservation, connection, and autoinjection (Voronin,

2025c), and derived from those laws a persistent reasoning graph, topology-aware navigation, and structural self-evaluation as architectural requirements for AI (Voronin, 2025d).

A subsequent synthesis reported a large factorial experimental program in which an Option K ontology bridge and topology retrieval did not produce a reproducible structural advantage on generic logic and constraint puzzles (Voronin, 2026). In the refined K=50 experiment, native SToE, chimera, music-control, similarity, and topology conditions converged within noise. That result was interpreted as evidence that the instrument was not isolating the structural prediction: the experiment placed an ontology around a transformer, but it did not test a live reasoning field in which failed hypotheses could become structurally relevant after a state change.

The present study changes the instrument. It tests the narrower prediction articulated in the architectural-gap paper: a system that conserves failed reasoning and preserves why it failed should benefit on nonlinear problems in which a previously rejected path becomes relevant again (Voronin, 2025d). The central comparison is not SToE vocabulary versus no vocabulary. It is observer-aware topological navigation versus strong semantic retrieval over the same stored reasoning artifacts, with explicit diagnostic controls for query-blind topology, removal of failures, randomized edges, removal of query relevance, and canonical-seed interventions.

The experiment also has an unusual methodological history that is reported rather than hidden. A predecessor v8 experiment failed to beat dense semantic memory overall and revealed, in postmortem, that the focal failed hypothesis was often reachable but mis-prioritized by query-blind graph-local navigation. V9 introduced observer/current-state-aware navigation but its first primary run was invalidated when a serializer bug was discovered: selecting four artifacts did not guarantee that four reached the model. V9.1 fixed only that measuring instrument, preserved the navigator and benchmark, and reran the study as a corrective replication. This sequence creates a useful test of whether the architectural change itself—not post-hoc tuning to a favorable score—accounts for the observed behavior.

2. Theoretical and Architectural Background

2.1 From SToE laws to a reasoning-memory architecture

The present experiment does not attempt to test the full SToE ontology. It tests an engineering consequence derived from three claims in the SToE research program. The Law of Information Conservation states that information points do not vanish but may lose connections; in AI terms, this motivates preserving failed hypotheses, contradictions, evaluations, and discarded paths rather than deleting them. The Law of Connection states that reachability depends on observer- or creator-established connections; in AI terms, this motivates navigation over the relational history of reasoning rather than similarity alone. The Law of Autoinjection places the reasoning system itself within the same information field; in architecture, this motivates representing the observer/current state and later evaluations as information points in the same graph (Voronin, 2025c, 2025d).

The architectural-gap derivation therefore proposed a directed persistent reasoning graph whose nodes are reasoning artifacts and whose typed edges encode relations such as generation, contradiction, evaluation, and failure provenance. The active context is not identical to the whole memory: it is a traversed subgraph selected from a field that continues to exist outside the model’s current token window (Voronin, 2025d). This distinction is important because conservation does not imply keeping every information point permanently active.

The v9.1 implementation extends this derivation with an explicit observer state. The current question, goal, active constraints, changed constraints, recent evidence, and reasoning position are represented by a `CurrentObserverStateIP`. State changes are first-class information points. A targeted task can therefore encode a path of the form `CurrentObserverStateIP → contains_change → StateChangeIP → invalidates → ConstraintIP ← rejected_by ← CandidateIP`. Reverse traversal of `rejected_by` is interpreted explicitly as

finding hypotheses previously rejected because of the invalidated constraint; it is not treated as a symmetric edge.

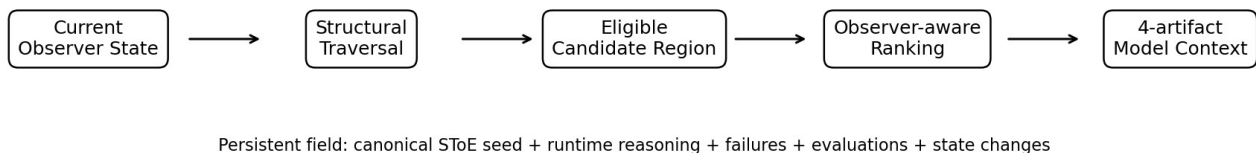


Figure 1. V9.1 information flow. Topology defines a bounded candidate region; observer/current-state signals rank candidates within that region before a fixed four-artifact model context is serialized.

2.2 Relation to prior SToE experiments

The prior Information Architecture article summarized a 3,900-call factorial program in which Option K supplied an ontology manifest and recorded mention-edges around a conventional LLM. The strongest K=25 observation did not replicate at K=50, and a chimera experiment showed no measurable difference between native SToE adjacency and music-shaped adjacency carrying the same SToE names (Voronin, 2026). Those experiments therefore provided no evidence that ontology topology itself improved generic puzzle solving.

The present design tests a different causal object. The graph is no longer only an ontology around the model. It is the runtime reasoning substrate: current state, changed constraints, rejected hypotheses, failed paths, evaluations, and the canonical SToE bootstrap field coexist in one traversable graph. The target phenomenon is also different. Instead of asking whether graph structure raises average performance on ordinary puzzles, the benchmark asks whether relational provenance can recover an old failure whose relevance is caused by a changed constraint even when lexical or semantic similarity points elsewhere. Thus the present study should be read as a test of the architectural prediction in Voronin (2025d), informed by the negative calibration reported in Voronin (2026), rather than as a reanalysis of the earlier Option K claim.

3. Research Questions and Frozen Predictions

The v9.1 corrective replication retained the v9 frozen success logic. The primary targeted comparison asked whether observer-aware SToE topology would solve more of the nine topology-targeted tasks than dense semantic retrieval. A second primary comparison asked whether observer-aware navigation would outperform a query-blind topology baseline, directly testing the architectural change motivated by the v8 postmortem.

The frozen predictions were:

- P1. Observer-aware topology should outperform query-blind topology on topology-targeted tasks.
- P2. Observer-aware topology should obtain at least one net additional targeted-task success over dense semantic retrieval.
- P3. At least 25% of observer-only targeted wins should show counterfactual dependence on a frozen graph-memory mechanism.
- P4. Observer-aware topology should not be more than two tasks worse than dense semantic retrieval on the nine negative-control tasks.

Secondary analyses compared observer-aware topology with BM25, success-only memory, failure removal, untyped model-visible relations, randomized edges, removal of the explicit state-change score, removal of query relevance, and three canonical-seed conditions. These comparisons were not treated as independent confirmation of the full SToE theory.

4. Methods

4.1 System and persistent field

V9.1 used a directed information graph containing both runtime reasoning artifacts and a byte-preserved canonical SToE seed. The seed contained 36 core information points, 113 typed relations, six categories, 36 expressions, and 32 non-empty descriptions. Its SHA-256 was b327db6dbce9981ed21561b9d1a857e2e3786f1391d41edc0381fead79e59868. Seed and runtime nodes occupied the same traversable field. The seed could be folded into a domain IP and unfolded losslessly, but it was never concatenated wholesale into the LLM prompt. Seed nodes could reach the model only through the same retrieval and memory budget applied to other eligible artifacts.

Runtime nodes represented current observer state, task evidence, hypotheses, failures/rejections, state changes, constraints, evaluations, and other reasoning artifacts. Each model-visible artifact carried a stable MEMORY_REF separate from its semantic content; internal IP/seed references were forbidden from satisfying exact-answer scoring. This separation was introduced to avoid the storage-ID/output-token ambiguity identified in earlier development runs.

4.2 Observer-aware navigator

Candidate generation was structural. Starting from CurrentObserverStateIP, the system traversed the directed field in permitted directions to depth four. Non-visible structural nodes could support paths but were filtered before ranking and top-k selection. The structural path score used a 0.65 distance decay, normalized by accumulated decay and penalized by path length so that longer paths could not win merely by accumulating positive edge weights.

Within the structurally generated candidate region, the observer-aware score combined structural relevance, bounded goal relevance, state-change relevance, provenance, failure relevance, distance, redundancy, and stale-locality penalties. In the frozen specification the coefficients were 0.25 structural + 0.15 goal + 0.30 state-change + 0.10 provenance + 0.10 failure relevance – 0.05 distance – 0.10 redundancy – 0.15 stale locality. Scores were machine-readable navigation evidence and were not exposed to the LLM.

The query-blind topology condition preserved a v8-style locality bias: it used normalized structural path score plus a fixed 0.50 bonus for the current-reasoning IP and did not use question, goal, active constraint, outcome, or answer content. It therefore served as a diagnostic of the architectural change rather than as a claim about the strongest possible graph retriever.

4.3 Benchmark

The frozen primary benchmark contained 18 distinct tasks grouped into nine families: planning, debugging, constraint satisfaction, legal-rule reasoning, scientific-hypothesis reasoning, code repair, hidden-rule reasoning, contradiction networks, and multi-agent planning. Each family contributed one topology-targeted task and one negative control. The family, rather than the raw task count, was the cluster unit for bootstrap inference.

Topology-targeted tasks shared a causal motif but differed in domain content: a candidate was previously rejected because of a constraint; later evidence invalidated or removed that constraint; the final question did not necessarily resemble the old candidate semantically. Negative controls were designed so that recent or directly similar evidence should remain sufficient and topology should not require reactivation of a prior failure. The benchmark was frozen before v9.1 corrective execution and retained the v9 benchmark hash 8c74a2af83c98d79ff33db99180052a5e54566ec9f555c19093bade4a8a43b62.

4.4 Experimental conditions

Fourteen frozen conditions were evaluated. All memory-bearing conditions selected four eligible artifacts. The principal baselines were no memory, success-only memory, BM25 retrieval, dense semantic retrieval,

typed semantic retrieval, and query-blind topology. The focal condition was OBSERVER_AWARE_STOE_TOPOLOGY. Diagnostic ablations removed the core seed, replaced it with a frozen offset-7 chimera, removed failed artifacts, masked relation labels after typed selection, randomized edge endpoints, zeroed the explicit state-change signal, or zeroed query relevance.

Condition	Operational difference
NO_MEMORY	No retrieved memory
SUCCESS_MEMORY	Success-only memory
BM25_MEMORY	Lexical BM25 retrieval
DENSE_SEMANTIC_MEMORY	Dense embedding retrieval
TYPED_SEMANTIC_MEMORY	Dense retrieval with typed artifact text
QUERY_BLIND_TOPOLOGY	Structural locality without observer relevance
OBSERVER_AWARE_STOE_TOPOLOGY	Full observer-aware topology + canonical seed
...WITHOUT_CORE_SEED	Full navigator, seed removed
...CHIMERA_SEED	Full navigator, seed adjacency rewired
...WITHOUT_FAILURES	Failed/rejected artifacts unavailable
...UNTYPED	Typed selection plan; relation labels masked to LLM
...RANDOMIZED_EDGES	Typed endpoints deterministically randomized
...WITHOUT_STATE_CHANGE_SIGNAL	Explicit numerical state-change component zeroed
...WITHOUT_QUERY_RELEVANCE	Goal/query relevance component zeroed

Table 1. Frozen v9.1 experimental conditions (abbreviated condition labels in the lower rows refer to OBSERVER_AWARE_TOPOLOGY/STOE prefixes in the source specification).

4.5 Model, execution, and information budget

All generation calls were local through Ollama 0.32.13. The frozen generation model was gemma4:26b, 25.8B parameters, Q4_K_M quantization, digest 5571076f3d70050487b26b341705799e0ab29b808164f90d20d4cf84f699d251. Dense retrieval used qwen3-embedding:0.6b, 1024 dimensions, Q8_0, digest ac6da0dfba84a81fdbfbaf330198c33cd77c4cdfc53e8bc50eb581914a15621d. Generation used temperature 0.0, top_p 0.9, top_k 40, seed 9417, an 8,192-token context, a 192-token output ceiling, JSON response format, and one permitted transport retry. There was one model call per task-condition primary cell.

The primary experiment contained $18 \times 14 = 252$ cells. All non-no-memory conditions used exactly four model-visible artifacts, subject to a common 5,000-character memory ceiling. Across the completed primary run there were zero provider errors, zero parse errors, and zero transport retries. Total recorded primary tokens were 147,965.

4.6 Why v9 was invalidated and what v9.1 changed

The first v9 primary run was not treated as valid confirmatory evidence. Post-run audit discovered that its 5,000-character serializer appended full memory blocks until the next block would exceed the limit and then executed a break. As a result, selecting four artifacts did not guarantee that four reached Gemma: one oversized seed artifact could remove itself and every later selected artifact. The recorded selected-artifact count therefore overstated actual model exposure in affected conditions.

V9.1 changed only post-selection serialization and exposure accounting. The 5,000-character total budget was retained. Metadata space was reserved for every selected artifact; remaining payload was deterministically distributed; oversized artifacts were individually truncated with a CONTENT_TRUNCATED marker; and the exact final memory string was parsed immediately before generation. If four eligible artifacts had been selected, the call was aborted unless four unique MEMORY_REF blocks were actually visible. The navigator, graph, seed, chimera, benchmark, model settings, condition definitions, statistics, success criteria, and counterfactual definitions remained frozen.

Fifty tests passed before execution. A full no-model dry run found zero exposure violations, zero memory-budget violations, and zero frozen-selection mismatches. In the completed primary run, all 234 memory-bearing cells exposed four selected artifacts and none exceeded 5,000 characters. The result is therefore a corrective replication of an outcome-known benchmark, not an independent confirmation.

4.7 Statistical analysis

The estimand was paired absolute success-rate difference. Confidence intervals were percentile intervals from 10,000 family-cluster bootstrap resamples with seed 9417. Two-sided exact McNemar/binomial tests were applied to discordant task pairs. The primary targeted comparison was observer-aware topology versus dense semantic retrieval. The frozen success criterion required all four conditions stated in Section 3; it did not require $p < .05$ for the dense comparison. Consequently, the formal preregistered verdict and conventional inferential uncertainty are reported separately.

4.8 Counterfactual mechanism tests

After completion of all primary cells, 18 successful observer-aware cases were replayed under five frozen interventions: removal of the focal failed IP, removal of the state-change IP, removal of the rejected_by relation class, randomization of critical edge endpoints, and removal of query ranking. Replays used the corrected serializer and the same exposure assertions. These are pipeline interventions: they change graph traversal and replacement context, so they support dependence on the implemented mechanism rather than minimal text-only causal claims.

5. Results

5.1 Primary condition performance

Observer-aware SToE topology solved all 18 primary tasks, including all nine topology-targeted tasks and all nine negative controls. The strongest conventional retriever overall was typed semantic memory at 15/18; dense semantic and BM25 each scored 14/18. On topology-targeted tasks, dense semantic solved 6/9, typed semantic 6/9, and BM25 5/9. Query-blind topology solved 0/9 targeted tasks, success-only memory 0/9, and the without-failures ablation 0/9.

Condition	Overall	Targeted	Controls	Focal retrieved
No memory	0/18 (0.0%)	0/9	0/9	0/18
Success-only	3/18 (16.7%)	0/9	3/9	4/18
BM25	14/18 (77.8%)	5/9	9/9	14/18
Dense semantic	14/18 (77.8%)	6/9	8/9	15/18
Typed semantic	15/18 (83.3%)	6/9	9/9	15/18
Query-blind topology	4/18 (22.2%)	0/9	4/9	4/18
Observer-aware SToE	18/18 (100.0%)	9/9	9/9	18/18
No core seed	16/18 (88.9%)	9/9	7/9	16/18
Chimera seed	13/18 (72.2%)	9/9	4/9	13/18
Without failures	4/18 (22.2%)	0/9	4/9	4/18
Untyped exposure	18/18 (100.0%)	9/9	9/9	18/18
Randomized edges	3/18 (16.7%)	3/9	0/9	3/18
No state-change scalar	18/18 (100.0%)	9/9	9/9	18/18
No query relevance	13/18 (72.2%)	9/9	4/9	13/18

Table 2. Primary v9.1 performance. OA = observer-aware. Focal retrieval denotes whether the task’s focal previously failed/relevant artifact appeared among the four selected artifacts.

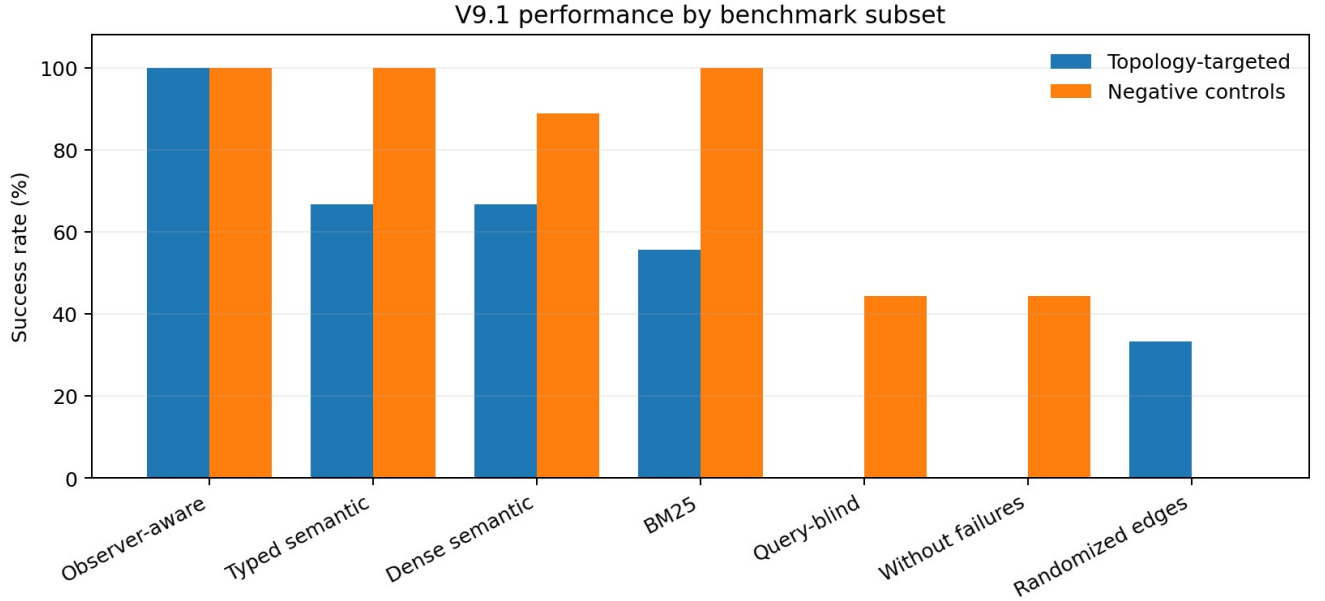


Figure 2. Success rates for key conditions on topology-targeted tasks and negative controls.

5.2 Primary paired comparisons

On the nine topology-targeted tasks, observer-aware topology scored 9/9 versus 6/9 for dense semantic retrieval, an absolute difference of +33.3 percentage points. The family-cluster bootstrap 95% CI was [0.0, 66.7] percentage points and the two-sided exact paired p-value was .25. The direction and magnitude meet the frozen criterion, but the interval touches zero and the exact test is not conventionally significant; the sample is therefore statistically imprecise.

The architectural comparison with query-blind topology was larger. Observer-aware topology scored 9/9 versus 0/9 targeted (difference +100 points; exact p=.003906) and 18/18 versus 4/18 overall (difference +77.8 points; family-cluster 95% CI [61.1, 94.4], exact p=.000122). This comparison directly tests the change motivated by the predecessor postmortem: making current observer state and changed constraints part of navigation rather than using graph-local recency alone.

Observer-aware topology also scored 9/9 targeted versus 0/9 without failures (exact p=.003906), and 9/9 versus 3/9 with randomized edges (difference +66.7 points; family-cluster 95% CI [33.3, 100.0], exact p=.03125). These results are consistent with both retained failed information and non-random connectivity contributing to the targeted mechanism.

Comparison	Observer	Comparator	Difference	95% cluster CI	Exact p
Dense semantic	9/9	6/9	+33.3 pp	[0.0, 66.7]	0.250000
Query-blind	9/9	0/9	+100.0 pp	[100.0, 100.0]	0.003906
Without failures	9/9	0/9	+100.0 pp	[100.0, 100.0]	0.003906
Randomized edges	9/9	3/9	+66.7 pp	[33.3, 100.0]	0.031250

Table 3. Selected frozen paired comparisons on the nine topology-targeted tasks.

5.3 Counterfactual dependence in observer-only wins over dense retrieval

Dense semantic retrieval failed three targeted tasks that observer-aware topology solved: primary-enzyme-control, primary-witness-chain, and primary-robot-capability. In all three cases the observer-aware trace contained the focal failed hypothesis plus an invalidates → rejected_by provenance path, all four selected artifacts were model-visible, and there was no internal-ID answer contamination.

Removing the focal failed IP changed success to failure in all three tasks. Removing the rejected_by relation class also changed success to failure in all three, as did randomizing the relevant edge class/endpoints. These interventions are not minimal single-token manipulations—the graph change alters downstream

retrieval—but they provide pipeline-level evidence that success depended on the conserved failed hypothesis and the structural route by which it became relevant.

Task	Primary outcome	Remove focal failed IP	Remove rejected_by class	Randomize critical edges
primary-enzyme-control	Observer correct; dense wrong	success → failure	success → failure	success → failure
primary-witness-chain	Observer correct; dense wrong	success → failure	success → failure	success → failure
primary-robot-capability	Observer correct; dense wrong	success → failure	success → failure	success → failure

Table 4. Clean topology-only wins over dense retrieval and corrected counterfactual interventions.

5.4 Canonical seed analyses

Under corrected equal-four-artifact exposure, the native canonical seed condition scored 18/18, the same observer-aware navigator without the core seed scored 16/18, and the frozen chimera seed scored 13/18. All three conditions were 9/9 on topology-targeted tasks; differences arose only on controls. Native versus no-seed differed by two tasks (exact $p=.5$), while native versus chimera differed by five (exact $p=.0625$). These are secondary, underpowered comparisons and do not establish a general benefit from the canonical SToE ontology.

Nevertheless, the seed was operational rather than decorative. Native selected 27 canonical-seed artifacts, nine of which were truncated under the fixed budget. Five selected runtime paths used seed IPs as unexposed bridge nodes. This demonstrates an architectural distinction anticipated by the field model: an information point can participate in navigation without necessarily consuming a model-visible context slot. The result is compatible with useful seed-mediated connectivity on some controls, but a larger blinded experiment is needed to separate native adjacency from benchmark-specific structure.

6. Discussion

6.1 What is supported

The strongest conclusion is not that SToE as a whole has been validated. The strongest conclusion is that a specific SToE-derived memory mechanism behaved as predicted on the problem class built to test it. When earlier failures were conserved, relational provenance was preserved, and the current observer/state participated in ranking structurally reachable candidates, the system recovered all nine targeted tasks. Query-blind topology recovered none; removing failures recovered none; randomizing edge endpoints reduced performance to 3/9.

This matters because the predecessor development result was negative in a diagnostically useful way. The earlier graph could often reach the focal failed node but systematically preferred the immediately preceding dead end at final decision time. V9 changed the architectural variable implied by that failure—observer/current-state-aware relevance—rather than merely increasing graph weights for the desired node. In v9.1, observer-aware and query-blind conditions selected different contexts on all 18 tasks; those changes improved binary correctness on 14 and harmed none. The 9/9 versus 0/9 targeted contrast therefore provides a direct mechanism-level test of the architectural correction.

6.2 What is not yet supported

The comparison with dense semantic retrieval is promising but not definitive. The +33.3-point targeted point estimate is based on only nine family clusters, with a 95% bootstrap interval that includes zero at its boundary and a two-sided exact p -value of .25. The frozen success criterion was intentionally mechanistic and required only a net targeted advantage plus counterfactual and control guards; meeting that criterion should not be confused with conventional statistical confirmation of superiority over dense retrieval.

The benchmark also favors a particular structural motif: a changed constraint invalidates the reason that an earlier hypothesis was rejected. The observer-aware navigator was explicitly designed to exploit invalidates

and reverse rejected_by paths. This makes the benchmark appropriate for mechanism validation, but it limits generalization. A strong architecture should eventually demonstrate benefit across unseen graph motifs, unseen task writers, different domains, and different generation models.

6.3 Relation to the earlier null Option K experiments

The present positive result does not reverse the earlier null finding reported in Information Architecture (Voronin, 2026). The two experiments target different mechanisms. Option K asked whether placing SToE ontology content and ontology topology around an LLM improved generic puzzle performance. The refined answer was no detectable structural effect. V9.1 asks whether a live reasoning graph containing failed hypotheses, state changes, and provenance can navigate back to a formerly rejected hypothesis when the reason for rejection changes. The latter phenomenon was not instantiated by the earlier instrument.

In this sense, the negative factorial program served as calibration. It ruled out a simpler claim—that SToE ontology topology by itself produces a robust puzzle-solving lift—and motivated a more direct test of the architectural requirement derived in The Architectural Gap (Voronin, 2025d). Scientific progress here comes from narrowing the claim, not from treating every negative result as support for the framework.

6.4 What the ablations do and do not mean

The untyped condition also scored 18/18, but it inherits the candidate selection plan produced by the typed navigator and masks relation labels only in model-visible context. It therefore shows that Gemma did not need to see the relation names after typed navigation had already selected the correct nodes; it does not show that relation semantics are irrelevant to navigation. Likewise, the without-state-change-signal ablation scored 18/18, but invalidates and rejected_by relations remained available through structural/provenance terms. The result isolates the numerical state-change bonus, not all state-change structure.

The without-query-relevance condition scored 13/18, with 9/9 targeted but 4/9 controls. This suggests that the targeted change-path motif was sufficiently strong to identify the focal failure structurally, while query relevance was important for avoiding irrelevant topology on control tasks. This pattern is consistent with a division of labor in which topology defines what is structurally reachable and observer relevance helps determine what matters now.

6.5 Corrective replication as a methodological constraint

The v9 serializer defect materially affected the first primary run and therefore had to invalidate it. V9.1 is stronger because it preserved the failed run, changed only serialization/accounting, verified that selected references remained identical in all 252 cells, and reran the frozen architecture. However, the benchmark outcomes were already known before the corrective replication. V9.1 therefore cannot be treated as an independent confirmation even though the principal navigator was unchanged. The appropriate next step is not another tuned version of the engine but a blind holdout using the frozen v9.1 architecture.

7. Limitations and Next Tests

First, the effective independent sample is nine family clusters. This is enough to expose large diagnostic contrasts such as 9/9 versus 0/9, but not enough to estimate the observer-versus-dense advantage precisely. A larger holdout should preregister a primary effect size and power target before outcomes are inspected.

Second, only one local generation model and one frozen model digest were used. Although deterministic generation parameters were specified, three v9-to-v9.1 correctness flips occurred even in cells whose exposure was unchanged, demonstrating residual nondeterminism in the local inference stack. Replication should include at least one second model family while keeping the memory controller fixed.

Third, equal artifact counts and a shared 5,000-character ceiling do not imply equal semantic information content. Different retrievers can select artifacts of different lengths and informational quality. Future designs could add token-normalized or matched-content controls without changing the core selection question.

Fourth, the counterfactual interventions are pipeline-level. Removing a node or relation changes what later retrieval supplies. Stronger causal isolation would compare minimally different model-visible contexts while holding all noncritical text constant, alongside the graph intervention.

Fifth, canonical-seed results are exploratory. Native 18/18 versus chimera 13/18 is suggestive but small and outcome-known, and the targeted subset was identical across seed modes. A dedicated native-seed experiment should use multiple degree-preserving chimeras or label permutations and a fresh benchmark to separate meaningful adjacency from general graph geometry.

The immediate next experiment should therefore freeze v9.1 unchanged and evaluate it on a new blind holdout generated independently of the navigator coefficients. The holdout should include several structural motifs beyond invalidated rejection: superseded rules, changed source reliability, revoked dependencies, delayed contradiction resolution, and multi-hop provenance where the useful old artifact is deliberately semantically distant from the final query. Conventional retrievers should receive the same stored content and the same model-visible budget.

8. Conclusion

This study tested a narrow prediction derived from the SToE research program: failed reasoning should remain available as structured information, and a reasoning system should be able to return to it when the current observer state makes it relevant again. In a frozen 18-task, 14-condition corrective replication, observer-aware topology solved 9/9 topology-targeted tasks, compared with 6/9 for dense semantic retrieval, 5/9 for BM25, 0/9 for query-blind topology, and 0/9 when failures were removed. Corrected counterfactuals showed pipeline-level dependence on the focal failed hypothesis and the rejected_by relation class in all three observer-only wins over dense retrieval.

The result is positive evidence for the implemented architectural mechanism, not for the entire Special Theory of Everything. The dense comparison remains underpowered, the benchmark is mechanism-targeted, and v9.1 is corrective rather than independent confirmation. The next scientific test is therefore straightforward: keep the architecture frozen, move to an unseen benchmark and additional model family, and ask whether the same observer-aware topological advantage survives without benchmark-specific design knowledge.

Data, Code, and Reproducibility

The complete v9.1 reproduction package is publicly available as a ResearchGate data record: Voronin, M. (2026), SToE v9.1 Observer-Aware Topological Memory - Reproduction Package, DOI: 10.13140/RG.2.2.22378.48327. The archive contains the frozen benchmark, canonical SToE seed, source code, tests, preregistered specification, pre- and post-run manifests, raw primary traces, primary results, counterfactual results, exposure audits, statistical analysis, and reproduction command. The completed raw primary-result SHA-256 is 1546d7f9950cbfbff869213eb0198f96d955e277cfbb5bf3ef6c648e0cd41503.

Generative AI Assistance Statement

Generative AI tools were used during software development, adversarial code/experiment auditing, and manuscript drafting. Experimental conditions, freezes, local model execution, integrity checks, and final scientific responsibility remained under the author's direction. The primary generation model evaluated in the experiment was the frozen local Gemma4:26b instance described in Methods; drafting assistance was not itself an experimental condition.

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Appendix A. Frozen Primary Benchmark Structure

The 18 tasks were organized as paired topology-targeted and negative-control tasks within nine independent families. Exact prompts and answers are contained in the frozen benchmark file in the public v9.1 reproduction package (Voronin, 2026; DOI: 10.13140/RG.2.2.22378.48327).

Family	Topology-targeted task	Negative-control task
planning	primary-canyon-supply	primary-planning-direct
debugging	primary-decoder-revival	primary-debug-recent
constraint satisfaction	primary-orchard-layout	primary-constraint-direct
legal rule	primary-harbor-rule	primary-legal-recent
scientific hypothesis	primary-enzyme-control	primary-science-direct
code repair	primary-vector-patch	primary-code-recent
hidden rule	primary-symbol-rule	primary-hidden-direct
contradiction network	primary-witness-chain	primary-contradiction-recent
multi agent planning	primary-robot-capability	primary-agent-direct

Table A1. Frozen benchmark task pairs by family.